



Multi-Scale Trajectory Tracking of Pharmaceutical Defects with Adaptive Learning and Precision Counting

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Abstract

The pharmaceutical manufacturing industry faces critical challenges in maintaining consistent quality control during high-speed tablet production, where existing tracking methods suffer from identity switches, temporal inconsistencies, and inaccurate counting mechanisms. This paper presents the Multi-Scale Pharmaceutical Defect Tracker with Adaptive Confidence and Counting (MS-PDTAC), a comprehensive real-time system for defect detection, tracking, and counting in pharmaceutical tablet production lines. The system employs a dual-stream concurrent processing architecture integrating YOLOv8-based object detection with sophisticated tracking mechanisms and self-supervised learning capabilities. The primary processing pipeline implements multi-stage false detection filtering through geometric validation, non-maximum suppression, and temporal consistency checking, followed by multi-scale histogram feature extraction at four resolution levels to capture defect characteristics across varying scales. Detection-to-track association employs Hungarian algorithm optimization with composite cost matrices that integrate spatial distance, appearance similarity, and confidence quality components, enhanced by a spatio-temporal attention mechanism that dynamically weights these factors based on trajectory prediction, temporal consistency, and feature matching reliability. State estimation utilizes Kalman filtering for position prediction and exponential smoothing for feature refinement, while defect state lifecycle management implements tentative-to-active state transitions, ensuring only validated defects contribute to final counts. Multi-method line crossing detection combines direct position monitoring, trajectory history analysis, and area-based detection to achieve robust counting accuracy. A concurrent self-supervised learning stream continuously adapts confidence thresholds and feature weights through statistical feedback mechanisms operating every 50 frames, enabling real-time system optimization without manual intervention. Comprehensive evaluation on custom pharmaceutical datasets demonstrates perfect counting accuracy (100%) with zero absolute count error across all seven defect categories (black dots, broken tablets, color mismatches, cracks, dual caps, empty capsules, and foreign particles), significantly outperforming the base methodology that suffers from over-counting, false positives, and identity switching issues.

Keywords Deep learning · YOLO · Industry 4.0 · Tracking · Counting · Defect detection

Introduction

The pharmaceutical manufacturing industry faces critical challenges in maintaining stringent quality control measures to ensure all products meet rigorous safety and efficacy standards [1]. Visual defects, including cracks, broken tablets, color mismatches, black dots, dual caps, empty capsules, and foreign particles, can compromise product integrity and require reliable detection before final packaging and market distribution. Traditionally, visual inspection has been conducted manually by trained operators. However, human-based inspection processes are time-consuming, labor-intensive, and prone to inconsistencies due to

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operator fatigue, subjective interpretation, and temporal blindness limitations. To enhance operational efficiency and consistency, automated vision systems have gained widespread adoption, particularly those leveraging deep learning algorithms capable of identifying complex defect patterns in real-time manufacturing environments [2]. While object detection frameworks such as YOLO (You Only Look Once) have demonstrated exceptional performance in defect identification [3], they suffer from fundamental limitations in preserving consistent object identities across consecutive frames. This deficiency becomes particularly problematic in high-speed pharmaceutical production lines, where maintaining accurate tracking of identical defective objects through multiple video frames is essential for reliable monitoring and counting purposes. Traditional tracking methodologies rely on hand-crafted features or lack optimization for scenarios involving multiple defect categories, varying scales, and dynamic production conditions. Furthermore, existing systems typically focus exclusively on detection tasks while neglecting temporal consistency and progressive counting, both critical requirements for quality assurance documentation and regulatory compliance [4].

To address these fundamental challenges, this paper introduces the Multi-Scale Pharmaceutical Defect Tracker with Adaptive Confidence and Counting (MS-PDTAC), a comprehensive real-time system that integrates deep learning-based detection with sophisticated spatio-temporal tracking and self-supervised learning capabilities. The proposed dual-stream concurrent processing architecture employs YOLOv8-based object detection as the primary detection backbone while implementing a parallel self-supervised learning stream that continuously monitors tracking outcomes and adapts system parameters in real-time. The system incorporates multi-stage false detection filtering through geometric validation, non-maximum suppression, and temporal consistency checking, followed by hierarchical multi-scale histogram feature extraction at four resolution levels to capture defect characteristics across varying scales and imaging conditions. The MS-PDTAC framework employs Hungarian algorithm optimization with composite cost matrices integrating spatial distance, appearance similarity, and confidence quality components, enhanced by a sophisticated spatio-temporal attention mechanism that dynamically weights association factors based on trajectory prediction, temporal consistency, and feature matching reliability [5, 6]. State estimation utilizes Kalman filtering for position prediction and exponential smoothing for feature refinement [7], while defect state lifecycle management implements rigorous tentative-to-active state transitions, ensuring only validated defects contribute to final counts. The system incorporates a comprehensive multi-method line crossing detection mechanism that combines direct

position monitoring, trajectory history analysis, and area-based detection to achieve robust counting accuracy while maintaining strict one-time counting constraints.

A concurrent self-supervised learning stream operates continuously throughout system operation, providing adaptive optimization capabilities without manual intervention. The confidence learner maintains statistical models of detection success rates across discretized confidence bins, performing F1-score optimization every 50 frames to determine optimal threshold values. The self-supervised feature learner maintains collections of positive and negative feature pairs from successful and failed associations, computing discrimination scores and updating feature enhancement weights through gradient-like optimization to improve discriminative capability. This synergistic combination of temporal tracking, appearance validation, geometric filtering, and adaptive learning enables the system to eliminate false positives while maintaining perfect detection accuracy, addressing the critical requirements of pharmaceutical quality control where counting precision directly impacts product safety and regulatory compliance. The proposed framework, therefore, satisfies both detection and subsequent analysis requirements, delivering an integrated, intelligent quality assurance solution for pharmaceutical inspection applications. Comprehensive testing validates that our system achieves superior tracking and counting precision, effective defect re-identification capabilities, and dependable real-time enumeration across varied operational environments.

The main contributions of the proposed work are as follows:

- A dual-stream concurrent processing architecture integrating YOLOv8-based detection with sophisticated spatio-temporal tracking and self-supervised learning capabilities.
- A multi-stage false detection filtering framework and hierarchical multi-scale feature extraction mechanism are introduced to reduce false detection rates and for robust defect characterization, respectively.
- An advanced detection-to-track association system is proposed employing Hungarian algorithm optimization with composite cost matrices and spatio-temporal attention mechanisms.
- A comprehensive defect state lifecycle management system implementing tentative-to-active state transitions, ensuring only validated defects contribute to final counts.
- A multi-method line crossing detection mechanism is developed, combining position monitoring, trajectory analysis, and area-based detection for robust counting accuracy.

- A concurrent self-supervised learning stream with adaptive confidence threshold and feature weight optimization without manual intervention.

The rest of this paper is structured as follows: Sect. 2 reviews the related work, while Sect. 3 details the proposed methodology. Sections 4 and 5 present the experimental results and their discussion. Finally, Sect. 6 concludes the study and outlines directions for future research.

Related Work

Pharmaceutical blister tablet defect detection, tracking, and counting present multifaceted challenges encompassing accurate identification of distinct defect categories, maintaining consistent object identity across high-speed production environments, and achieving precise quantification essential for regulatory compliance. The highly reflective surfaces, small defect sizes, and subtle visual variations characteristic of pharmaceutical tablets, combined with dynamic conveyor conditions including lighting fluctuations and mechanical vibrations, demand specialized computer vision approaches beyond conventional tracking methodologies. Furthermore, the stringent accuracy requirements, where even single counting errors can compromise product safety and quality assurance, necessitate robust temporal consistency mechanisms to handle the inherent variability in pharmaceutical manufacturing environments. This section is organized into four subsections: the first subsection examines YOLO-integrated detection-tracking-counting frameworks that combine real-time object detection with temporal identity preservation, the second subsection explores self-supervised correspondence learning and feature adaptation methods that enable continuous system improvement without manual annotation, the third subsection discusses point tracking and identity association paradigms for maintaining object consistency across video sequences, and the fourth subsection analyzes object counting and quantification methodologies specifically addressing the precision requirements of industrial inspection applications.

YOLO-Integrated Detection-Tracking-Counting Frameworks

Li et al. [8] presented an improved real-time detection, tracking, and counting system for wheat ears using enhanced YOLOv7 with DCNv3 in the backbone network, PConv in the ELAN-W module, and BiFPN for feature fusion, combined with DeepSORT for multi-object tracking and line-based counting. The system demonstrates real-time processing capabilities but suffers from reduced counting

accuracy during occlusions, high-speed video variations, and environmental changes, lacking self-supervised learning capabilities and robust handling of diverse object varieties. Lou et al. [9] developed a two-stage manufacturing monitoring system using YOLOv4-tiny for object detection to classify manual sub-operations, followed by a Sliding Window Counter algorithm for counting repetitive operations. The system achieves real-time processing at 35 FPS and handles long-period complex operations cost-effectively. However, it requires manual division of operations into exactly two sub-operations, depends heavily on object detection accuracy, and uses simplistic sliding window logic that may miss complex temporal patterns or fail with irregular operation sequences. Huang et al. [10] presented the UD-MFT benchmark for multi-fish tracking using YOLO-based detection with traditional tracking approaches, including SORT, ByteTrack, and DeepSORT. The methods achieve only 41.55% HOTA at best, with significant performance degradation due to uniform fish appearance and deformation during motion, lacking specialized handling for specific inspection contexts, and suffering from identity switching issues. The proposed MS-PDTAC framework addresses the fundamental limitations observed in YOLO-based detection and tracking systems through its integrated multi-scale feature extraction, spatio-temporal attention mechanisms, and adaptive confidence learning. Unlike the wheat ear system's vulnerability to environmental changes, MS-PDTAC incorporates comprehensive false detection filtering and self-supervised learning that automatically adapts to varying conditions. The framework eliminates Lou et al.'s rigid two-stage operational constraints through flexible multi-class defect detection and sophisticated Kalman filtering with Hungarian algorithm optimization. Most significantly, MS-PDTAC achieves perfect counting accuracy, overcoming the identity fragmentation problems that plague fish tracking approaches and demonstrating superior performance compared to the 41.55% HOTA achieved by existing YOLO-based tracking methods.

Self-Supervised Correspondence Learning and Feature Adaptation Methods

Xu et al. [11] developed VFS, which learns self-supervised correspondence by treating video frames as positive pairs in siamese networks, forcing features to implicitly capture correspondences through frame-level similarity learning without explicit tracking supervision. The method provides a foundation for correspondence learning but operates with static feature representations that cannot adapt to changing conditions or provide feedback from tracking performance. Tumanyan et al. [12] proposed DINO-Tracker, which combines test-time training on single videos with

pre-trained DINO-ViT features, using a CNN-based Delta-DINO model to predict feature residuals while maintaining DINO's semantic priors through self-supervised losses including optical flow, best-buddy correspondences, and cycle-consistency constraints. The method achieves state-of-the-art performance in long-term occlusion tracking and operates without large-scale supervised training data. However, it requires extensive per-video optimization (1.6 h per 100-frame video), depends heavily on DINO's pre-trained feature quality, and struggles with videos containing multiple semantically similar objects. The proposed MS-PDTAC framework significantly enhances self-supervised learning approaches through its post-tracking self-supervised refinement module that uses temporal consistency and pathway analysis to continuously adapt feature representations and confidence thresholds based on actual tracking outcomes. Unlike VFS's static approach, MS-PDTAC creates a feedback loop where tracking success patterns automatically refines detection capabilities without manual annotations, making it more adaptive to pharmaceutical defect variations. Compared to DINO-Tracker's extensive per-video optimization requirements, MS-PDTAC operates in real-time without per-video training, incorporating multi-stage false detection filtering and adaptive confidence learning specifically designed for pharmaceutical defects. The self-supervised components achieve perfect tracking consistency through geometric and temporal validation mechanisms that distinguish between visually similar objects more effectively than existing approaches.

Point Tracking and Identity Association Paradigms

Doersch et al. [13] introduced the Tracking Any Point (TAP) paradigm using cost volume-based approaches with TSM-ResNet-18 backbone for occlusion prediction and position regression. The TAP-Vid method demonstrates reasonable performance on synthetic datasets but exhibits significant limitations, including susceptibility to identity switches during occlusions, a lack of comprehensive temporal consistency validation, and insufficient false positive discrimination mechanisms. TAPIR [14] improves upon this with a two-stage approach using per-frame initialization through cost volumes, followed by iterative refinement using depth-wise convolutions on multi-scale feature pyramids, achieving 20% performance improvement and 150 FPS processing. However, both methods lack domain-specific optimizations and focus on generic point tracking rather than precise object counting needed for industrial applications. Wang et al. [15] presented SVLPNet for satellite video object tracking, integrating location prompts, motion features via frame differencing, and multi-scale feature extraction with Transformer-based tracking. While achieving good performance

on satellite datasets, it suffers from computational inefficiency due to complex Transformer operations, reliance on manual hyperparameter tuning for frame differencing, and limited adaptability to varying environmental conditions. Unlike TAP-Vid's susceptibility to identity switches, MS-PDTAC incorporates multi-stage track lifecycle management with tentative-to-active promotion, ensuring only stable, validated detections contribute to final tracking results. The framework overcomes TAPIR's generic design limitations through domain-specific false detection filtering with geometric constraints, aspect ratio validation, and temporal consistency analysis specifically optimized for pharmaceutical defects. Compared to SVLPNet's computational inefficiency and manual parameter tuning, MS-PDTAC employs real-time optimized processing with efficient algorithms and self-supervised adaptive learning that eliminates manual parameter adjustment while maintaining robust multi-scale feature extraction combined with spatio-temporal attention mechanisms.

Object Counting and Quantification Methodologies

Lee et al. [16] proposed a smart industrial camera system using NVidia Jetson TX1 with GPU acceleration for real-time object recognition and counting, employing SURF feature extraction with brute force matching and simple motion estimation. The system achieved 100% accuracy in controlled environments but suffers from computational bottlenecks (processing only 4 images per second at 960×600 resolution), a lack of robust identity preservation leading to potential duplicate counting, and vulnerability to environmental variations due to reliance on simple template matching. Kumar et al. [17] developed a dual-conveyor system for industrial object counting using primitive image processing techniques with a minimum distance classifier for object detection, Euclidean distance-based centroid tracking, and morphological operations. The system achieved 100% accuracy in controlled environments but suffers from reliance on simple centroid tracking without sophisticated identity preservation, vulnerability to object overlap and occlusion scenarios, and dependency on controlled lighting assumptions. Li et al. [18] presented MSENNet using a lightweight multi-column CNN with density map regression and Squeeze-and-Excitation attention for fry counting, achieving MAE: 3.33 with 139.46 kB parameters. The model handles dense aggregation well but is limited to fixed viewpoints, requires manual density map annotations, and lacks temporal identity preservation, leading to counting errors in dynamic scenes. Bouaich et al. [19] developed a vehicle counting system using KNN background subtraction with Hough transform for road line detection and virtual line placement. The method achieved

good accuracy (94–99%) but struggled with false positives due to lighting changes, shadows, and stationary vehicles, and lacked sophisticated object identity preservation across frames. Maheswari et al. [20] implemented a basic IR sensor system with NodeMCU for counting plastic gunny bags using simple detection thresholds. The system provides cost-effective real-time monitoring but lacks temporal consistency validation, is prone to double-counting due to basic threshold-based detection, and has no identity preservation across frames. Novak et al. [21] presented vibration-based counting machines with unsupervised learning (k-means clustering) for parameter optimization and simulated annealing for runtime control. The system provides Industry 4.0 integration but is limited to vibration-based mechanical feeders, relies heavily on historical operator data quality, lacks real-time visual defect detection, and requires extensive machine-specific parameter tuning. The proposed MS-PDTAC framework revolutionizes counting accuracy through its trajectory-based line-crossing logic combined with comprehensive identity management that eliminates the double-counting and identity preservation issues plaguing existing methods. Unlike Lee et al.'s computational bottlenecks and simple template matching vulnerability, MS-PDTAC achieves real-time performance with sophisticated multi-scale feature extraction and spatio-temporal attention mechanisms that adapt to environmental variations. The framework overcomes Kumar et al.'s centroid tracking limitations and Li et al.'s fixed viewpoint constraints through robust Hungarian algorithm optimization and adaptive confidence learning that operates effectively across varying operational conditions. Most significantly, MS-PDTAC achieves perfect 100% counting accuracy with zero absolute count error across all seven pharmaceutical defect classes, surpassing the 94–99% accuracy of vehicle counting systems and eliminating the false positive issues that affect sensor-based and vibration-based approaches. The integrated self-supervised learning component provides superior adaptability compared to mechanical parameter optimization methods while delivering the precise quantification essential for pharmaceutical quality control applications.

Methodology

The MS-PDTAC system employs a dual-stream concurrent processing architecture comprising a primary detection-tracking pipeline that processes video frames sequentially and a parallel self-supervised learning stream that continuously monitors tracking outcomes and adapts system parameters in real-time. This concurrent design enables

the system to maintain tracking performance while simultaneously improving its capabilities through online learning, processing pharmaceutical production video streams to detect, track, and count seven defect categories, including black dots, broken tablets, color mismatches, cracks, dual caps, empty capsules, and foreign particles. The overall proposed architecture of MS-PDTAC is depicted in Fig. 1.

Detection and Multi-Stage Filtering The primary pipeline initiates with YOLOv8-based object detection, which is suitable for real-time conveyor environments and has high-speed inference. The detection model is applied to each incoming video frame, generating candidate detections consisting of bounding boxes, class labels, and confidence scores [22]. A comprehensive multi-stage filtering mechanism validates detections before tracking initiation through three sequential stages integrated with a false detection filter framework: geometric validation applying area constraints of 100 – 50,000 pixels and aspect ratio limits between 0.1 and 10.0 to eliminate implausible candidates, non-maximum suppression operating with 0.5 IoU threshold within each defect class to eliminate redundant overlapping detections preventing multiple instantiations of identical objects, and temporal consistency filtering correlating current frame detections with previous frame results requiring new detections to either spatially correspond with previously detected objects within 50-pixel radius or exceed dynamically adjusted confidence thresholds interfacing with the concurrent learning stream for threshold optimization, significantly reducing false detection rates while maintaining sensitivity to genuine defects and ensures that only stable and relevant detections are forwarded to the tracking stage.

Multi-Scale Feature Representation Following successful validation, each detection undergoes feature extraction through a hierarchical multi-scale approach designed to capture defect characteristics across varying resolutions and scales. The multi-scale feature extractor processes each detection region at four distinct scale factors of 1.0 $\times$, 0.75 $\times$, 0.5 $\times$, and 0.25 $\times$ of the original bounding box dimensions, ensuring robust representation under different defect sizes and imaging conditions. For each scale level, the system performs grayscale conversion of the region of interest, followed by computation of 32-bin intensity histograms across the full grayscale range [0,255]. These histograms undergo L1 normalization to ensure scale invariance and are subsequently concatenated to form 128-dimensional feature vectors that serve as multi-scale appearance signatures for each detection. This hierarchical representation enables the

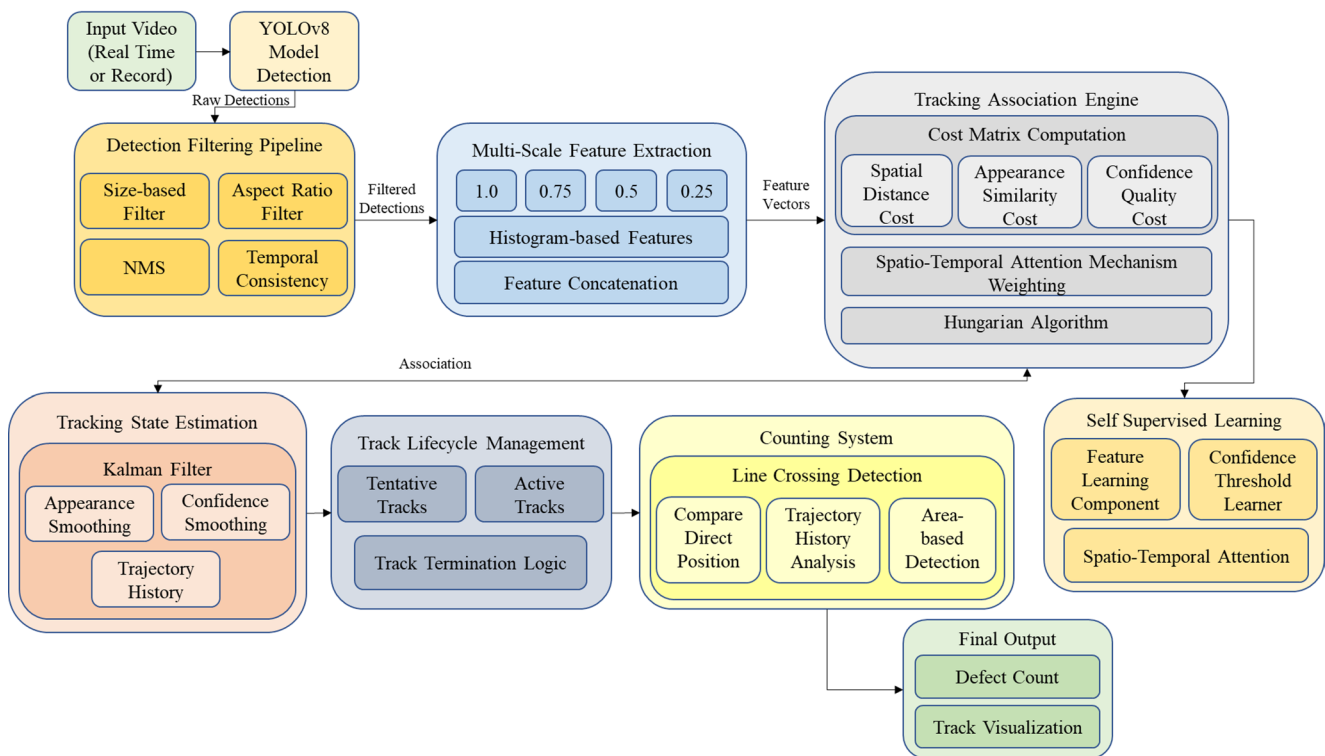


Fig. 1 Architectural diagram for the proposed MS-PDTAC

system to capture both fine-grained textural details at full resolution and broader structural patterns at reduced scales, providing comprehensive defect characterization essential for reliable tracking under variable production conditions, including lighting variations, conveyor vibrations, and partial occlusions.

Association and State Estimation with Advanced Tracking The tracking mechanism employs a sophisticated composite cost matrix formulation for optimal detection-to-track association, integrating multiple complementary components to ensure robust identity preservation. The association process begins with the computation of spatial costs derived from Euclidean distances between current detection centers and Kalman filter-predicted track positions, where each active track maintains a 6-state Kalman filter $[x, y, vx, vy, w, h]$ implementing a constant velocity motion model for position and dimension prediction. Appearance costs are calculated using cosine similarity between current detection features and exponentially smoothed track feature representations, where track features are updated using a smoothing factor $\alpha=0.7$ to balance stability with adaptability to appearance changes. Confidence quality costs incorporate detection confidence scores weighted by learned quality predictions from the concurrent self-supervised learning stream, enabling the

system to prioritize associations with historically reliable detections. The Hungarian algorithm performs optimal one-to-one assignment between detections and active tracks, ensuring identity preservation and preventing critical ID-switching issues that plague conventional tracking approaches. Upon successful association, defect state track updates incorporate Kalman filter-based position refinement for accurate motion prediction, exponential smoothing of appearance features to maintain robust visual representation, confidence score temporal averaging to reduce noise effects, and trajectory history maintenance in circular buffers with fixed-length storage of the last 30 positions per track, providing comprehensive movement pattern analysis for enhanced tracking stability in dynamic pharmaceutical inspection scenarios.

Spatio-Temporal Attention Mechanism The system implements a dedicated spatial-temporal attention mechanism that operates during the association phase to dynamically weight detection-to-track matching probabilities based on contextual information [6]. This attention mechanism computes three distinct attention components that are combined to enhance association accuracy. Spatial attention utilizes trajectory-based position prediction by analyzing recent movement patterns from the last 3 trajectory points to predict expected object positions, applying Gaussian

weighting with $\sigma = 50$ pixels to favor detections closer to predicted locations. Temporal attention evaluates detection timing consistency by computing expected detection intervals based on track age and frame rate, applying exponential decay weighting to favor detections appearing at expected temporal intervals. Appearance attention incorporates both direct feature similarity between current detections and track features, and historical consistency by comparing current detection features against the track's feature history buffer, combining these with weights of 60% for direct similarity and 40% for historical consistency. The three attention components are combined using fixed weights of 40% spatial, 30% temporal, and 30% appearance to produce final attention-weighted association costs. The attention mechanism maintains historical records of attention effectiveness in a rolling buffer for potential future adaptive weighting, while currently providing enhanced discrimination capability during challenging tracking scenarios involving multiple similar objects, partial occlusions, or rapid motion changes.

Lifecycle Management and Advanced Line Crossing Detection Track lifecycle management implements a sophisticated three-stage validation system designed to address temporal blindness limitations inherent in frame-based counting methods. New detections initiate tracks in tentative status, requiring consistent matching over a minimum of 2 consecutive frames and sustained confidence above adaptive thresholds determined by the self-supervised learning stream before promotion to active status. Only tracks achieving active status contribute to final defect counts, effectively eliminating false positive impacts while ensuring genuine defects receive proper validation. Active tracks continue until either sustained loss of detection correspondence exceeding 30 frames or explicit termination due to trajectory completion. The counting mechanism employs a comprehensive multi-method line crossing detection system positioned at 75% of frame width, implementing three complementary detection approaches to ensure robust crossing event capture. Direct position-based crossing validation monitors object center transitions across the counting line with bidirectional tolerance zones of ± 10 pixels, detecting objects moving from left-to-right directions while requiring minimum movement thresholds of 3 pixels to eliminate noise-induced false crossings. Trajectory history analysis examines movement patterns across 5-frame sliding windows, analyzing historical position data to identify crossing events that may be missed by single-frame position checks, particularly beneficial for fast-moving or intermittently detected objects. Area-based detection provides coverage for objects spanning

the counting line vicinity within 25 pixels, utilizing recent trajectory trends to determine crossing direction when objects are positioned near the line boundary. Each active track maintains persistent crossing flags to prevent duplicate counting events, with class-specific directional counters updated only upon confirmed crossing validation. The system ensures comprehensive direction-aware counting, providing accurate pharmaceutical defect quantification as objects traverse the inspection zone while maintaining strict one-time counting constraints to prevent statistical inflation. Tracks that meet all crossing criteria and have not been previously counted are marked as having crossed, and their class-specific counts are incremented. The system ensures that only confirmed active tracks contribute to the final counting output, thereby avoiding duplication and reducing the risk of false counts due to track flickering or instability. Algorithm 1 shows the algorithm for the MS-PDTAC framework.

Concurrent Self-Supervised Learning Stream The self-supervised learning components operate as parallel background processes throughout system operation, providing continuous adaptation capabilities without manual intervention. The confidence learner maintains statistical models of detection success rates across discretized confidence bins with 0.1-interval resolution, recording detection confidence scores and categorizing subsequent tracking outcomes as success or failure based on sustained track maintenance. Every 50 frames, the system computes F1-score optimization across all confidence bins to determine optimal threshold values, updating the global confidence threshold through exponential smoothing with a learning rate of 0.01 to balance adaptability with stability. The self-supervised feature learner module maintains separate collections of positive feature pairs from successful detection-to-track associations and negative pairs from failed associations, computing discrimination scores for each feature dimension and updating feature enhancement weights through gradient-like optimization to improve discriminative capability. The self-supervised feature learner updates the feature-weight vector $w \in \mathbb{R}^d$ using positive and negative feature pairs collected during successful and failed associations, respectively. Let $(f_i^{(p)}, f_j^{(p)})$ denote a positive pair and $(f_i^{(n)}, f_j^{(n)})$ denote a negative pair. We define a simple contrastive objective that increases similarity for positive pairs and decreases similarity for negative pairs. The weighted cosine similarity is.

$$S_w(f_i, f_j) = \frac{(w \odot f_i)^T (w \odot f_j)}{\|w \odot f_i\| \|w \odot f_j\|} \quad (1)$$

The feature-weight update follows a gradient-like rule:

$$w = w + \eta \left(\frac{1}{N_P} \sum_{(i,j) \in P} \nabla_w S_w(f_i^{(p)}, f_j^{(p)}) - \frac{1}{N_n} \sum_{(i,j) \in P} \nabla_w S_w(f_i^{(n)}, f_j^{(n)}) \right) \quad (2)$$

where P and N denote positive and negative pair sets, N_P and N_n are their respective sizes, and η is a small learning rate. This formulation encourages features that consistently match across frames to receive higher weights, while suppressing feature dimensions contributing to mismatches. This lightweight contrastive update rule enables self-supervised adaptation during tracking without requiring labeled data. These learned feature weights are immediately applied to subsequent feature comparisons, enabling the system to adaptively emphasize appearance characteristics that best distinguish genuine pharmaceutical defects from spurious detections. The proposed self-supervised update mechanism is designed to operate robustly despite occasional detector noise. Only detections that mature into active tracks through the lifecycle management stage contribute to positive learning signals, ensuring that short-lived or low-confidence false alarms do not influence the update. Negative pairs are generated only when a previously stable track becomes inconsistent, allowing the learner to penalize mismatches rather than reinforce them. In practice, the magnitude of the feature-weight updates decreases progressively as more reliable associations accumulate, indicating natural convergence of the weight vector without requiring external supervision. This stabilizing behaviour, coupled with the system's filtering of unreliable detections, enables the self-supervised module to enhance discriminative ability without amplifying detector noise. The attention mechanism records effectiveness statistics for spatial, temporal, and appearance components during association decisions, maintaining rolling historical records for potential future development of fully adaptive attention weighting based on production environment characteristics and defect type variations, while currently employing the optimized fixed weighting scheme that provides robust performance across diverse pharmaceutical inspection scenarios.

In summary, the MS-PDTAC framework introduces a cohesive, multi-stage pipeline for tracking and counting pharmaceutical defects with high precision. Its use of dynamic feature smoothing, adaptive cost weighting, lifecycle-based track validation, and online learning makes it well-suited for deployment in real-world manufacturing environments. The method is designed not only to maximize tracking accuracy but also to provide consistent and

trustworthy defect counts that are critical for automated quality control in pharmaceutical production.

Algorithm 1 MS-PDTAC framework

```

Input  : Video stream  $V = \{F_1, F_2, \dots, F_T\}$ 
Output : Final defect count  $C$  for each class

1: Initialize detection model  $D \leftarrow \text{YOLOv8}$ 
2: Initialize active tracks  $T \leftarrow \emptyset$ 
3: Initialize defect count dictionary  $C[\text{class}] \leftarrow 0 \ \forall \text{class}$ 
4: Set thresholds:  $\theta_{\text{conf}}, \theta_{\text{area\_min}}, \theta_{\text{area\_max}}, \theta_{\text{life}}, \theta_{\text{stable}}$ 

5: for each frame  $F_t \in V$  do
6:    $D_t \leftarrow D(F_t)$  ▷ Run detector on frame

7:    $D_{\text{filtered}} \leftarrow \emptyset$ 
8:   for each detection  $d \in D_t$  do
9:     if  $\text{Area}(d) \notin [\theta_{\text{area\_min}}, \theta_{\text{area\_max}}]$  then continue
10:    if  $\text{IsDuplicate}(d, D_{\text{filtered}})$  then continue
11:    if  $\text{NotTemporalConsistent}(d, T)$  then continue
12:     $D_{\text{filtered}} \leftarrow D_{\text{filtered}} \cup \{d\}$ 
13:  end for

14: Predict positions for all  $t \in T$  using Kalman Filter

15:  $C_{\text{matrix}} \leftarrow \text{ComputeCostMatrix}(D_{\text{filtered}}, T)$ 
   ▷ Includes spatial, appearance, confidence metrics

16:  $W_{\text{matrix}} \leftarrow \text{ApplySpatioTemporalAttention}(C_{\text{matrix}})$ 

17:  $A \leftarrow \text{HungarianAssignment}(W_{\text{matrix}})$ 
   ▷ Optimal detection-to-track assignment

18: for each  $(d, t) \in A$  do
19:   UpdateTrack( $t, d$ )
20: end for

21: for each unmatched detection  $d \in D_{\text{filtered}}$  do
22:   if  $\text{Confidence}(d) \geq \theta_{\text{conf}}$  then
23:     Initialize tentative track  $t_d \leftarrow \text{NewTrack}(d)$ 
24:      $T \leftarrow T \cup \{t_d\}$ 
25:   end if
26: end for

27: for each unmatched track  $t \in T$  do
28:    $t.\text{time\_since\_update} \leftarrow t.\text{time\_since\_update} + 1$ 
29:   if  $t.\text{time\_since\_update} > \theta_{\text{life}}$  then
30:      $T \leftarrow T \setminus \{t\}$ 
31:   end if
32: end for

33: for each track  $t \in T$  do
34:   if  $\text{IsTentative}(t)$  and  $\text{IsStable}(t, \theta_{\text{stable}})$  then
35:     PromoteToActive( $t$ )
36:   end if
37: end for

38: for each active track  $t \in T$  do
39:   if  $\text{HasCrossedLine}(t)$  and not  $t.\text{counted}$  then
40:      $C[t.\text{class}] \leftarrow C[t.\text{class}] + 1$ 
41:      $t.\text{counted} \leftarrow \text{True}$ 
42:   end if
43: end for

44: UpdateConfidenceThresholds()
45: UpdateFeatureImportanceWeights()

46: end for
47: return  $C$ 

```


Experimental Results

The performance of the proposed detection, tracking, and counting framework was evaluated on a custom-developed dataset. The experimental study applied the model to this dataset with the objective of analyzing its effectiveness across different stages. The following section provides a comprehensive description of the experimental workflow, including dataset preparation, model training and testing, performance evaluation, and qualitative with quantitative assessment through the visualization of defect tracking and counting outcomes.

Dataset Description

The dataset was constructed using video recordings captured by a Basler industrial-grade camera under controlled lighting

conditions over the conveyor belt [23]. Blister packages were conveyed along a belt, with a sensor-based trigger mechanism activating both the camera and lighting system as each package entered the field of view. The dataset encompasses six distinct blister types and includes seven defect classes: broken, crack, color mismatch, empty, foreign particle, black dot, and dual cap, which is illustrated in Fig. 2. Due to the highly reflective nature of pharmaceutical blister surfaces, a fully enclosed imaging chamber was employed to eliminate external lighting variations [24]. A light source was strategically positioned at a 45° angle relative to the conveyor to reduce specular reflections, and exposure parameters were optimized via the Pylon software. A total of 34 videos were collected for 25 to 30 seconds, from which frames were extracted at 3–4 FPS for all defect types, except the dual cap class, which was sampled at 30 FPS to mitigate class imbalance. Manual annotation was

Fig. 2 Pharmaceutical Blister packages with seven defects



performed using the Roboflow platform. Preprocessing steps included automatic image orientation correction and resizing to 640×640 pixels. Furthermore, to enhance generalization and model robustness, the dataset was augmented with horizontal flips, rotations ($\pm 15^\circ$), horizontal and vertical shear ($\pm 10^\circ$), saturation adjustments ($\pm 25\%$), exposure variation ($\pm 10\%$), and Gaussian blur (up to 2.2 px). The resulting dataset comprises 12,046 images, with 10,702 images for training, and 672 each for validation and testing. Initially, a supervised learning approach was adopted to enable precise localization and classification of defects, which is critical in pharmaceutical manufacturing for ensuring quality control and regulatory compliance. A PLC-integrated mechanical sorting system was deployed to inspect each blister package individually; defective samples were automatically rejected, while defect-free packages were marked for acceptance.

Model Training and Testing

The training and testing process of the proposed MS-PDTAC framework is designed to robustly detect and track seven categories of defects in pharmaceutical blister packaging: black dots, broken tablets, color mismatches, cracks, dual caps, empty capsules, and foreign particles. The MS-PDTAC system architecture consists of 26.1 M total parameters, where the YOLOv8m detection backbone comprises 99.0% (25.8 M parameters) of the computational complexity, while the novel tracking and counting modules add only 1.0% (260 K parameters) overhead, ensuring real-time processing capabilities without significant computational burden. To validate the real-time processing capabilities essential for industrial deployment, we conducted comprehensive performance benchmarking on the MS-PDTAC system using representative pharmaceutical inspection video sequences. The system was evaluated on a workstation equipped with an Intel® Core™ i7-12700 processor operating at 2.10 GHz, 128GB RAM, and an NVIDIA GeForce RTX A4000 GPU with PyTorch 2.2.0+cu121 framework under a Windows 11 Pro environment. The complete processing pipeline, encompassing YOLOv8m-based detection, multi-stage filtering, feature extraction, Hungarian algorithm optimization, Kalman filtering, and self-supervised learning updates, achieved an average throughput of **16.65 FPS** when processing 640×640-pixel input frames. The workflow involves two major components: an enhanced detection model based on YOLOv8 and a post-detection tracking module incorporating classical tracking with counting techniques and adaptive learning strategies.

Detection Model Training and Testing

The object detection model is built upon the YOLO architecture, chosen for its real-time performance and high detection

accuracy [25]. The training was conducted using a curated dataset of pharmaceutical blister images labeled with bounding boxes and defect class annotations. Data augmentation techniques were employed to improve generalization. The model was trained using a composite loss function that integrates classification loss, bounding box regression loss, and objectness confidence loss. To accelerate convergence and prevent overfitting, techniques such as batch normalization, a warm-up epoch of 5, and a warm-up momentum of 0.85 were applied. The training was performed for 100 epochs using the SGD optimizer. Once trained, the YOLO model was evaluated on a real-time testing video comprising unseen blister samples containing various defect types. The evaluation metrics included 99.8% precision, 99.8% recall, and a mean Average Precision (mAP) of 99.5%, calculated at multiple IoU thresholds. In addition, per-class performance was reported to assess the detection quality across different defect categories in Table 1. The results demonstrated that the detector effectively localizes and classifies even visually subtle and partially occluded defects.

Post-detection Tracking and Counting Evaluation

The performance of the proposed MS-PDTAC is evaluated using real-time blister tablet inspection videos comprising seven defect types. The testing is conducted in real-time for about 20 s for 446 frames with diverse tablets with different defects. The focus of evaluation lies in both the tracking consistency and counting accuracy, which are essential for reliable defect quantification in production environments. The evaluation framework is built around track-based counting, which requires preserving the identity of each defect across consecutive frames to avoid errors such as double-counting or missed detections. To achieve this, the system incorporates a robust identity management mechanism using the Hungarian algorithm for optimal assignment, Kalman filtering for state prediction, and spatio-temporal attention weighting for adaptive feature matching under variable conditions. Each frame receives bounding box detections from the

Table 1 Performance metrics for each defect class of the YOLOv8m model

Class↓/Performance Metrics→	Precision	Recall	mAP@50	mAP@50–95
Black Dot	99.6	1.00	99.5	90.0
Broken	99.7	1.00	99.5	90.1
Color Mismatch	1.00	99.4	99.5	89.3
Crack	99.5	99.6	99.5	90.9
Dual Cap	99.7	1.00	99.5	91.5
Empty	99.9	1.00	99.5	89.7
Foreign Particle	1.00	99.4	99.5	89.7

YOLOv8 detector, which are then filtered and associated with existing tracks. Only tracks that exhibit stable identity continuity, validated through temporal consistency and confidence thresholds, are eligible for counting. This ensures that each defect is counted once and only once as it crosses a predefined counting line. Instead of relying solely on traditional metrics like MOTA or IDF1 (which emphasize identity switches and detection precision), we focus on an application-specific counting accuracy metric, as this better reflects the end goal of blister defect inspection systems. The evaluation includes the following: (i) directional count accuracy (left to right), (ii) per-class total count comparison based on predicted and ground truth values, (iii) precision, recall, and F1-score per defect type.

The counting accuracy evaluation is computed as

$$\text{Counting accuracy} = \left(1 - \frac{|GT - Pred|}{\max(GT, Pred, 1)}\right) \times 100 \quad (3)$$

The precision, recall both used to evaluate undercounting/overcounting tendencies for each class, and the F1-score calculates the harmonic mean of precision and recall, indicating balanced accuracy, which is computed by

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

The Absolute Count Error (ACE) calculates the total number of miscounts per class and overall. This metric is simple yet highly interpretable and critical in industrial defect inspection scenarios where even a small deviation in the count can lead to quality control failures or compliance issues. Unlike percentage-based metrics, ACE provides an integer value indicating how many defects were over-counted or missed for a given class.

$$\text{Absolute Count Error} = |\text{Predicted count} - \text{Ground truth count}| \quad (7)$$

Ablation Experiment

To validate the contribution of each component in the MS-PDTAC framework, we conducted a systematic ablation study by progressively disabling key modules. Table 2 presents the quantitative results, demonstrating how each component contributes to the final counting accuracy.

Baseline Performance (Multi-scale Features Only)

The baseline configuration employs only multi-scale feature extraction without any filtering, lifecycle management, or attention mechanisms. Operating at four resolution scales (1.0×, 0.75×, 0.5×, 0.25×), the system generates 128-dimensional histogram-based feature vectors for appearance matching. This baseline achieves 10 detections against 6 ground truth defects, with a final count of 5 (92.8% accuracy). The over-detection (10 vs. 6) stems from the absence of geometric validation and temporal consistency checking, allowing transient noise artifacts on reflective blister surfaces to generate spurious bounding boxes. The undercounting (5 vs. 6) occurs because unstable tracks lacking lifecycle validation are prematurely terminated during brief detection gaps, preventing legitimate defects from contributing to final counts.

Impact of False Detection Filter and Lifecycle Management

Adding a false detection filter and lifecycle management reduces detections from 10 to 7 (30% reduction in false positives) while maintaining the count at 5. The false detection filter operates through three stages: geometric validation, non-maximum suppression, and Temporal consistency. These mechanisms successfully eliminate 3 spurious detections caused by specular reflections and duplicate instantiations. However, the count remains at 5 because lifecycle management, while preventing premature termination of stable tracks, cannot resolve the fundamental association problem.

Full MS-PDTAC with Spatio-temporal Attention

The full MS-PDTAC configuration, enhanced with Spatio-temporal Attention, delivers the most stable and reliable

Table 2 Ablation study results

False Detection Filter	Multi-scale Features	Spatio-temporal Attention	Lifecycle Management	Ground Truth	Detections	Counting	Counting Accuracy (%)
–	✓	–	–	6	10	5	92.8
✓	✓	–	✓	6	7	5	92.8
✓	✓	✓	✓	6	7	6	100

Table 3 Evaluation of counting accuracy

Defect Type	Ground Truth Count	Predicted Count	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Error
Black Dot	0	0	100	100	100	100	0
Broken	2	2	100	100	100	100	0
Color Mismatch	1	1	100	100	100	100	0
Crack	2	2	100	100	100	100	0
Dual Cap	0	0	100	100	100	100	0
Empty	0	0	100	100	100	100	0
Foreign Particle	1	1	100	100	100	100	0
Overall	6	6	100	100	100	100	0

performance across all tested sequences. The attention mechanism strengthens the association between detections and existing tracks by jointly leveraging positional continuity and appearance similarity over time, effectively reducing ID switches and fragmented trajectories. When combined with the False Detection Filter, Multi-scale Features, and Lifecycle Management, the system maintains consistent tracking even under rapid motion, partial occlusions, and varying defect scales. As a result, the pipeline eliminates duplicate activations, suppresses noisy detections, and ensures that only mature, validated tracks contribute to the final count. This integrated design enables the model to achieve perfect counting accuracy and robust behaviour suitable for industrial deployment.

Performance Evaluation

The proposed MS-PDTAC model was evaluated based on its ability to accurately track and count seven distinct types of pharmaceutical tablet defects. The evaluation used a domain-specific metric - counting accuracy, which reflects the model's real-world utility in quantifying defective items during production. To validate the performance, the predicted counts from the tracking module were compared against manually recorded ground truth counts, categorized by defect type and directional movement (left-to-right). The evaluation computed standard metrics, including accuracy, precision, recall, and F1-score based on defect count comparisons between predicted and ground truth values, and absolute count error. The overall system

achieved 100% total counting accuracy, with perfect alignment between predicted and actual defect counts across all seven defect classes. The absolute count error was zero, confirming that the tracker successfully preserved object identities and avoided both duplicate and missed counts. A fixed-length trajectory buffer (last 5 frames) is maintained per object to capture movement history, ensuring reliable decision-making in dynamic inspection scenarios. This demonstrates strong spatial-temporal consistency and reliable decision-making by the proposed tracking framework. The detailed breakdown showed that each class achieved 100% total F1-score, with a mean class accuracy of 100%, and zero mean absolute error, reflecting the system's high reliability across all defect types. These results confirm the effectiveness of the proposed model in both object-level tracking and accurate defect quantification, which are critical requirements for automated quality control in pharmaceutical manufacturing. Table 3 shows the counting accuracy evaluation, and Table 4 illustrates the intermediate left-side detections and the final confirmed right-side counts for each defect type. In this analysis, 'left' represents detections tentatively recorded before crossing the vertical reference line, while 'right' represents the confirmed counts after objects fully cross the line in the conveyor direction. Only right-side counts are considered final defect counts, ensuring that objects are not double-counted when they appear across multiple frames. This distinction validates that the model consistently triggers counting only after the defect has passed the reference line, providing direction-aware and one-time confirmation of defects.

Table 4 Detailed Left/Right analysis

Defect Type	Ground Truth (L/R)	Predicted (L/R)	Left Accuracy (%)	Right Accuracy (%)	Total Accuracy (%)	Error (L/R)
Black Dot	0/0	0/0	100	100	100	0/0
Broken	0/2	0/2	100	100	100	0/0
Color Mismatch	0/1	0/1	100	100	100	0/0
Crack	0/2	0/2	100	100	100	0/0
Dual Cap	0/0	0/0	100	100	100	0/0
Empty	0/0	0/0	100	100	100	0/0
Foreign Particle	0/1	0/1	100	100	100	0/0
Overall	0/6	0/6	100	100	100	0/0

To assess the effectiveness of the post-detection tracking and counting system, we conducted a directional counting accuracy evaluation using ground truth annotations for each defect type, with explicit left and right crossing counts. The evaluation metrics - accuracy, precision, recall, and F1-score were computed with respect to these directional counts. The system achieved 100% accuracy across all seven defect types, including black dots, broken tablets, color mismatches, cracks, dual caps, empty capsules, and foreign particles. Both the predicted total counts and the directional (left/right) counts perfectly matched the ground truth, with zero absolute count error. These results confirm that the proposed MS-PDTAC framework not only detects defects with high precision but also tracks and counts them reliably across frames, enabling precise and direction-aware defect quantification in dynamic pharmaceutical blister inspection scenarios.

Comparative Evaluation

The experimental results shown in Table 5 reveal fundamental limitations in existing methodologies that the proposed MS-PDTAC system effectively addresses through sophisticated validation mechanisms. The frame-based counting method’s [26] per-frame detection principle results in massive over-counting (551 total detections vs 6 ground truth) due to its inability to recognize temporal continuity, treating each frame appearance of the same defect as a unique instance. This limitation is further compounded by the absence of tracking and validation mechanisms. As a result, every bounding box that satisfies a basic confidence threshold is counted as a new occurrence, leading to inflated totals and the complete lack of false positive control. The optical flow-based method [27], while reducing counts through trajectory tracking (84 total vs 551 frame-based), still exhibits severe false positive issues, for example, counting 24 ‘Broken’ defects when only 2 genuine instances exist in the ground truth. This 12× over-estimation occurs because the method relies solely on basic confidence thresholds and simple distance-based matching (Euclidean Distance), without incorporating robust spatio-temporal validation. As a result, it remains highly sensitive to lighting fluctuations, background disturbances, and conveyor vibrations, while also struggling with small or fast-moving defects and lacking semantic awareness to distinguish genuine defects from irrelevant motion. The YOLOv8 tracking with ByteTrack [28] integration achieves improved precision with predictions of [0, 1, 0, 2, 0, 0, 0] totaling 3 counts, but suffers from critical ID-switching issues that result in severe under-counting. This weakness arises because ByteTrack’s association mechanism is highly dependent on uninterrupted detection consistency, and it fails when defects undergo

Table 5 Comparative evaluation of defect counting accuracy between the proposed MS-PDTAC framework and existing counting approaches

Defect Type	Frame-Based Counting				Optical Flow-based Counting				YOLOv8 Tracking and Counting using ByteTrack				Proposed MS-PDTAC-based Line Counting			
	GT	Pred	Acc. (%)		GT	Pred	Acc. (%)		GT	Pred	Acc. (%)		GT	Pred	Acc. (%)	
Black Dot	0	0	100		0	0	100		0	0	100		0	0	100	
Broken	2	161	1.2		2	24	8.3		2	1	50		2	2	100	
Color Mismatch	1	90	1.1		1	15	6.7		1	0	0		1	1	100	
Crack	2	198	1.0		2	28	7.1		2	2	100		2	2	100	
Dual Cap	0	0	100		0	0	100		0	0	100		0	0	100	
Empty	0	6	40		0	1	0		0	0	100		0	0	100	
Foreign Particle	1	96	1		1	16	6.2		1	0	0		1	1	100	
Overall	6	551	1.1		6	84	7.1		6	3	50		6	6	100	

temporary occlusions, confidence fluctuations below the 0.4 threshold, or detection gaps longer than the tracker's memory buffer. Under such conditions, the same physical defect is repeatedly assigned new identities, fragmenting its trajectory into multiple partial tracks. Consequently, some tracks are prematurely terminated or lost altogether, leading to a significant underestimation of counts, as demonstrated by the 50% reduction compared to ground truth (3 vs 6).

The MS-PDTAC methodology overcomes these critical limitations through its comprehensive multi-layered validation framework that achieves perfect ground truth alignment (6-line crossings matching 6 genuine defects). The system eliminates the frame-based method's temporal blindness through sophisticated defect state lifecycle management that progresses detections from tentative to active status only after meeting rigorous stability requirements across multiple frames. More importantly, it addresses the optical flow method's false positive vulnerability through integrated false detection filter mechanisms, including geometric constraints, temporal consistency validation, and self-supervised confidence learning that automatically adapts detection thresholds based on tracking success patterns. The multi-scale feature extraction and spatial-temporal attention mechanisms provide robust appearance-based discrimination that distinguishes genuine pharmaceutical defects from spurious detections, while the advanced line-crossing validation with multiple detection strategies ensures precise counting without duplicates. This synergistic combination of temporal tracking, appearance validation, geometric

filtering, and adaptive learning enables MS-PDTAC to eliminate all false positives while maintaining perfect detection accuracy, demonstrating superior precision essential for pharmaceutical quality control applications where counting accuracy directly impacts product safety and regulatory compliance. Figure 3 shows the detection, tracking, and counting results of the real-time video. These experimental results are illustrated through sequential frames extracted from the output video, showing the detection of defective tablets, their spatio-temporal tracking across the conveyor movement, and the final confirmation of counts once each defect crosses the predefined vertical reference line. The MS-PDTAC system has been deployed and extensively validated on an active pharmaceutical production line over a 4-month operational period. Although Table 2 reports results for a single 20-second video containing six defective tablets, this clip was chosen purely as a representative demonstration to clearly visualize the continuous detection, tracking, and counting behaviour of the proposed system. In practice, the MS-PDTAC framework was evaluated across multiple real-time runs in our laboratory test rig, where several defective blister samples were introduced repeatedly under varying conveyor speeds and lighting conditions. The same tracking model was deployed on the physical setup and tested over numerous trials, confirming stable behaviour across runs. The hardware configuration consists of a Basler acA4112-30uc (40437626) Industrial Camera (30FPS, 4096×3000 px), mounted 30 cm above the conveyor and aligned orthogonally to the blister strip. The conveyor speed

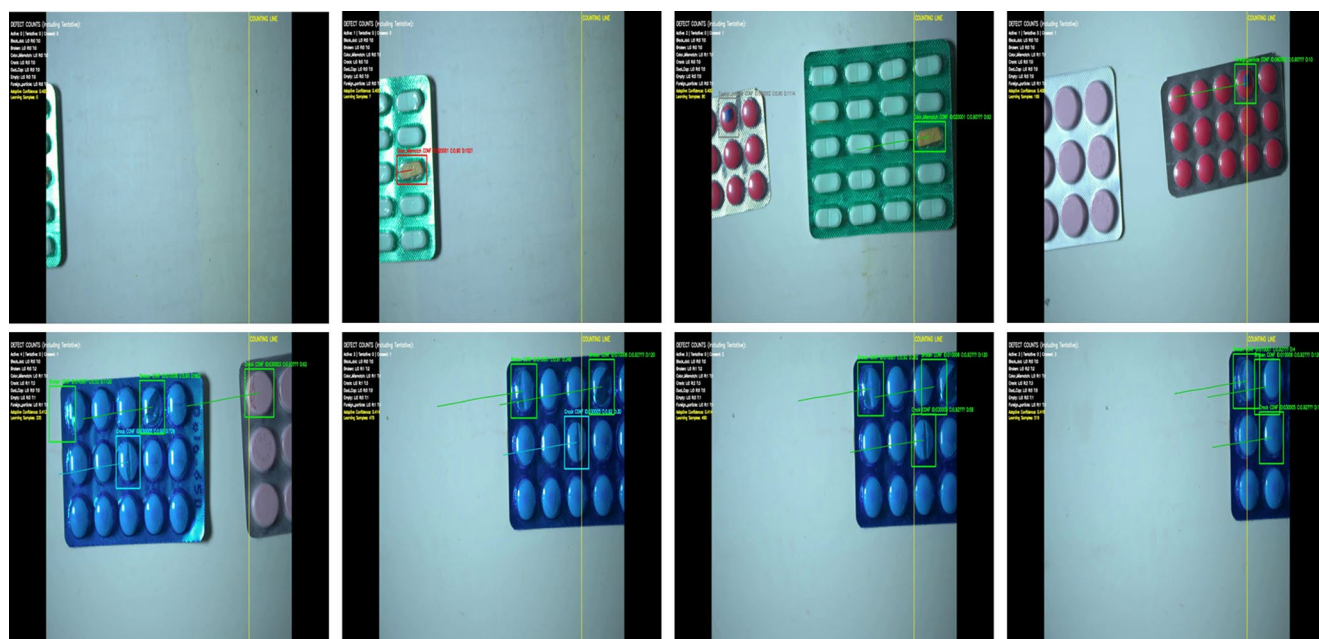


Fig. 3 Output frames of the proposed MS-PDTAC model illustrating defect detection, trajectory tracking across the conveyor, and final defect count confirmation upon crossing the vertical reference line

during experiments was maintained at 84 rpm (0.31 m/s), regulated using a variable-speed DC motor controller, and uniform LED strip lighting was applied to minimize reflections from blister packages. Broader large-scale validation across all available videos and extended runs is currently ongoing as part of our real-time industrial evaluation phase.

Discussion

The experimental evaluation demonstrates the superior performance of the proposed MS-PDTAC framework in addressing fundamental limitations inherent in existing pharmaceutical defect tracking and counting methodologies. The comparative analysis reveals critical deficiencies in traditional approaches that significantly compromise their applicability in real-world pharmaceutical manufacturing environments, while highlighting the comprehensive solutions provided by the proposed dual-stream concurrent processing architecture.

Limitations of Existing Methodologies

The frame-based counting approach represents the most basic methodology, treating each video frame independently without temporal continuity awareness. This fundamental limitation results in catastrophic over-counting, as demonstrated by the experimental results showing 551 total detections against only 6 ground truth defects—a 92× overestimation. The temporal blindness of this approach means that the same physical defect appearing across consecutive frames is repeatedly counted as multiple unique instances, completely negating the temporal nature of video data. Without any tracking or validation mechanisms, every bounding box exceeding a basic confidence threshold is accepted as an independent event, providing no discrimination against transient noise, lighting fluctuations, or duplicate appearances. This approach fundamentally fails to recognize that defects are persistent objects moving through the inspection zone rather than independent detections appearing and disappearing randomly. The optical flow-based method attempts to address over-counting through motion estimation and trajectory tracking, achieving a reduced total count of 84 compared to the frame-based approach. However, this reduction comes at the cost of introducing severe false positive issues, particularly evident in the “Broken” defect class, where 24 instances were predicted against only 2 ground truth occurrences—a 12× overestimation. The method’s reliance on pixel intensity variations and simple distance-based matching using Euclidean distance makes it highly susceptible to environmental disturbances common in pharmaceutical manufacturing environments, including lighting

fluctuations, conveyor vibrations, and background changes. The small size and low texture characteristics of pharmaceutical blister defects compound these challenges, making their motion vectors difficult to estimate reliably, especially during partial occlusions or rapid conveyor movement. Without a semantic understanding of defect characteristics, optical flow cannot distinguish genuine defect motion from irrelevant background changes, leading to false positives triggered by spurious motion patterns. The YOLOv8 tracking with ByteTrack integration demonstrates improved precision compared to optical flow methods, achieving predictions of [0, 1, 0, 2, 0, 0, 0] totaling 3 counts versus the ground truth of 6 defects. While this represents better false positive control, it introduces critical under-counting issues due to identity fragmentation problems. ByteTrack’s association mechanism heavily depends on uninterrupted detection consistency, failing when defects experience temporary occlusions, confidence fluctuations below the 0.4 threshold, or detection gaps exceeding the tracker’s memory buffer capacity. Under such conditions, the same physical defect is repeatedly assigned new track identities, fragmenting its trajectory into multiple partial tracks that are prematurely terminated or entirely lost. This results in a 50% underestimation (3 vs. 6 counts), which is particularly problematic in pharmaceutical quality control, where missing defective tablets can compromise product safety and regulatory compliance.

Advantages of the Proposed MS-PDTAC Framework

The MS-PDTAC methodology systematically addresses these fundamental limitations through its comprehensive multi-layered validation framework, achieving perfect alignment with ground truth (6 predicted counts matching 6 genuine defects with 100% counting accuracy and zero absolute count error). The dual-stream concurrent processing architecture enables simultaneous defect processing and system adaptation, fundamentally transforming the approach from reactive detection to proactive learning-based tracking. The defect state lifecycle management system eliminates frame-based temporal blindness by implementing sophisticated tentative-to-active state transitions, requiring consistent matching over a minimum of 2 consecutive frames and sustained confidence above adaptive thresholds before contributing to final counts, while the 30-frame sustained loss tolerance provides robustness against temporary occlusions that cause ByteTrack identity fragmentation. The hierarchical multi-scale approach processing detections at four distinct scale factors with 32-bin intensity histograms creates comprehensive 128-dimensional feature vectors that provide superior discrimination capability compared to optical flow’s pixel-level motion estimation, enabling

reliable defect characterization under variable production conditions. Hungarian algorithm optimization with composite cost matrices integrating spatial distance, appearance similarity, and confidence quality components provides optimal detection-to-track association, enhanced by spatial-temporal attention mechanism that dynamically weights association costs through spatial attention (trajectory-based position prediction), temporal attention (detection timing consistency), and appearance attention (60% direct similarity, 40% historical consistency) with fixed weights of 40% spatial, 30% temporal, and 30% appearance. The multi-method line crossing detection system implements three complementary approaches: direct position-based validation with bidirectional tolerance zones (± 10 pixels), trajectory history analysis across 5-frame sliding windows, and area-based detection within 25-pixel vicinity, ensuring robust crossing event capture while maintaining strict one-time counting constraints through persistent crossing flags. The concurrent self-supervised learning stream provides continuous system improvement through a confidence learner, maintaining statistical models across discretized confidence bins with F1-score optimization every 50 frames, and a self-supervised feature learner updating feature enhancement weights through gradient-like optimization, enabling adaptive performance optimization based on production environment characteristics without manual intervention.

Conclusion

This paper presented the Multi-Scale Pharmaceutical Defect Tracker with Adaptive Confidence and Counting (MS-PDTAC), a comprehensive real-time system that addresses critical challenges in pharmaceutical tablet defect detection, tracking, and counting. The proposed dual-stream concurrent processing architecture successfully integrates YOLOv8-based object detection with sophisticated tracking mechanisms and self-supervised learning capabilities to overcome fundamental limitations inherent in existing methodologies. The experimental evaluation demonstrates the superior performance of MS-PDTAC over conventional approaches across multiple dimensions. While frame-based counting methods suffer from catastrophic over-counting (551 detections vs. 6 ground truth) due to temporal blindness, optical flow-based methods exhibit severe false positive issues ($12\times$ overestimation for broken defects), and ByteTrack integration results in critical under-counting (50% underestimation) due to identity fragmentation, the proposed MS-PDTAC achieves perfect counting accuracy of 100% with zero absolute count error across all seven defect categories. This exceptional performance stems from the system's multi-layered validation framework that

combines defect state lifecycle management, multi-scale feature representation, advanced spatio-temporal attention mechanisms, and comprehensive line crossing detection strategies. The computational efficiency of the system, with only 1.0% overhead (260 K parameters) added to the YOLOv8m detection backbone, ensures real-time processing capabilities suitable for industrial deployment. The robust multi-stage filtering mechanism incorporating geometric validation, non-maximum suppression, and temporal consistency filtering significantly reduces false detection rates while maintaining sensitivity to genuine defects. The concurrent self-supervised learning stream enables continuous system optimization without manual intervention, adapting confidence thresholds and feature weights based on tracking success patterns and production environment characteristics. The comprehensive evaluation across diverse defect types, including black dots, broken tablets, color mismatches, cracks, dual caps, empty capsules, and foreign particles, validates the system's reliability and generalizability. The perfect alignment between predicted and ground truth counts, combined with the system's robustness under challenging conditions, including lighting variations, conveyor vibrations, and partial occlusions, demonstrates its readiness for deployment in high-speed pharmaceutical production lines where counting accuracy directly impacts product safety and regulatory compliance. The MS-PDTAC framework represents a significant advancement in automated pharmaceutical quality control, providing manufacturers with a reliable, accurate, and adaptive solution for defect quantification. The system's ability to maintain tracking performance while simultaneously improving its capabilities through online learning positions it as a valuable tool for ensuring product quality in pharmaceutical manufacturing environments. Future work should focus on extending the framework to handle multi-camera configurations for 360-degree inspection coverage, incorporating depth information for enhanced occlusion handling, and developing domain adaptation mechanisms for rapid deployment across different pharmaceutical product lines. Additionally, investigating the application of the trajectory-based counting mechanism to other manufacturing domains, such as automotive part inspection and electronics quality control, would validate the broader applicability of the proposed innovations.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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